Bhavana Mehta

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Education

University of Pennsylvania
Ph.D. & M.S. in Computer and Information Science — *2019 – 2025*

Nirma Institute of Technology
B.Tech., Electronics & Communication Engineering — *2014 – 2018*

Selected Publications

ZAYA1-8B Technical Report *(arXiv, May 2026)*
Bhavana Mehta et al. (in collaboration with Zyphra)

Adaptive Sharding in Untrusted Environments *(SIGMOD' 26)*
Bhavana Mehta, Nupur Baghel, Mohammad J Amiri, Ryan Marcus, Boon Thau Loo

Towards Full Stack Adaptivity in Permissioned Blockchains *(VLDB '24)*
Chenyuan Wu, Mohammad J Amiri, Haoyun Qin, **Bhavana Mehta**, Ryan Marcus, Boon Thau Loo

Towards Adaptive Fault-Tolerant Sharded Databases *(VLDB '23)*
Bhavana Mehta, Neelesh C A, Prashanth Iyer, Mohammad J Amiri, Boon Thau Loo, Ryan Marcus

AdaChain: A Learned Adaptive Blockchain *(VLDB '23)*
Chenyuan Wu, **Bhavana Mehta**, Mohammad J Amiri, Ryan Marcus, Boon Thau Loo

Human Action Recognition using Good Features and Multilayer Perceptron Network *(ICCSP '17)*
J. Talukdar, **Bhavana Mehta**

Experience

Zyphra
Member of Technical Staff, Performance Engineering — *Dec 2025 – Present*

- **Training Infrastructure (Aegis & Zookeeper):** Led job monitoring, restart orchestration, and checkpoint management for large-scale GPU training, including production 192-node BMOE 128k mid-train runs with failure recovery.
- **Checkpointing & Operator Tooling:** Fixed optimizer-state validation failures, migrated checkpoint logic from Aegis to Zookeeper, and built Slack controls for checkpoint sync and rekick toggles.
- **Inference Cloud & Production Ops:** Benchmarked DFlash/Eagle3-MLA, backported DFlash to vLLM v0.18.1 for **1.6–1.8x throughput wins**, built Perfetto/B2 profiling workflows, and handled storage/disk incidents.

University of Pennsylvania
Research Fellow, Distributed Systems Lab (Advisor: Prof. Boon Thau Loo) — *2019 – 2025*

- **AdaChain (RL for Systems):** Built a **reinforcement learning**/contextual-bandit framework for adaptive blockchain execution, including ledger state featurization, online model training, and decentralized architecture switching.
- **Marlin (Algorithmic Sharding):** Led adaptive sharding for untrusted distributed systems; benchmarked KaHyPar-based centralized resharding and decentralized consensus protocols, improving throughput by **over 40%** under adversarial skew.
- **Fault-Tolerant Systems:** Designed health monitoring/state replication for 70B+ LLM serving failover and built a DPDK TCP migration prototype with Microsoft Research/MIT, achieving $\sim 4\,\mu s$ migration latency and a **12x improvement**.

Bluespec Inc.
Design Engineer (prev. Intern) — *2018 – 2019*

- Tuned RISC-V CPU pipeline and **branch predictor (statistical modeling)** for compute-intensive workloads, **improving IPC by 25%**; concurrently built a Jenkins/Docker-based CI/CD pipeline that reduced hardware release cycles from two weeks to three days.

IIT Madras, RISE Lab
Research Intern — *Summer 2017*

- Designed a novel SRT Radix 4 algorithm for high-speed division; this design was integrated into India's first **open-source RISC-V processor** (SHAKTI), improving parallel quotient selection.

Skills

Languages:	Python, C++/C, SQL, Bash
Machine Learning:	Reinforcement Learning, Contextual Bandits, PyTorch, Scikit-learn
Distributed Systems:	Raft/Paxos, PBFT/HotStuff, 2PC, Redis, KaHyPar, YCSB
ML Infrastructure:	Distributed training orchestration, Checkpoint validation & recovery, Slurm, NFS/NVMe, Prometheus, Backblaze B2, vLLM, Perfetto
Systems & DevOps:	Linux, Docker, Git, Jenkins, CI/CD, HPC, CloudLab, Performance Optimization
Networking:	DPDK, Kernel Bypass, Low-Latency Networking